HW03

ENAE694: Spacecraft Communications

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Problem 1

Antenna Gain and Beamwidth *(30 pts.)*

A circular spacecraft reflector antenna operates at $f = 7.50\,\text{GHz}$, with diameter $D = 0.80\,\text{m}$. Calculate peak gain directly from diameter and frequency using the formula below, which assumes aperture efficiency 0.55.

a) Calculate peak antenna gain $[G]$ in dBi and full half-power beamwidth (HPBW) in degrees. *(10 pts.)*

b) Double the diameter while keeping frequency and efficiency fixed. Find the change in the gain, the new gain, and the new HPBW. *(10 pts.)*

c) Return to the original antenna. Double the frequency from 7.50 to 15 GHz while keeping diameter and efficiency fixed. Find the change in gain and the new HPBW. Explain why this gain increase alone does not establish the change in received power for the complete link. *(10 pts.)*

Use D in meters and f in GHz. The gain constant includes efficiency 0.55 and all unit conversions; do not apply efficiency again. HPBW is the full width in degrees. Its reflector rule is approximate and depends on illumination, not efficiency alone.

$$[G] = 17.81 + 20\log_{10}(D) + 20\log_{10}(f)$$

$$\text{HPBW} \approx \frac{21}{fD}\ \text{degrees}$$

a) Substituting $D = 0.80\,\text{m}$ and $f = 7.50\,\text{GHz}$ (as plain numbers in meters and GHz) directly into the gain formula:

$$\begin{aligned}[G] &= 17.81 + 20\log_{10}(0.80) + 20\log_{10}(7.50)\\ &= 17.81 + (-1.9382) + 17.5012\\ &= 33.3730\,\text{dBi}\end{aligned}$$

and the full half-power beamwidth is

$$\text{HPBW} \approx \frac{21}{fD} = \frac{21}{(7.50)(0.80)} = \frac{21}{6.00} = 3.50^\circ$$

```
D = 0.80 m, f = 7.50 GHz
20 log10(D) = -1.9382 dB
20 log10(f) = 17.5012 dB
[G]  = 33.3730 dBi ~ 33.37 dBi
HPBW = 3.5000 deg ~ 3.50 deg
```

$$[G] = 33.37\,\text{dBi}, \quad \text{HPBW} = 3.50^\circ \quad \blacksquare$$

b) With $D' = 2D = 1.60\,\text{m}$ and f unchanged, only the diameter term changes:

$$\Delta[G] = 20\log_{10}\left(\frac{D'}{D}\right) = 20\log_{10}(2) = 6.0206\,\text{dB}$$

$$[G'] = [G] + \Delta[G] = 33.3730 + 6.0206 = 39.3936\,\text{dBi}$$

$$\text{HPBW}' \approx \frac{21}{(7.50)(1.60)} = \frac{21}{12.00} = 1.75^\circ$$

Doubling the aperture quadruples its area (gain $\propto D^2$, so $+6.02\,\text{dB}$) and halves the beamwidth (HPBW $\propto 1/D$).

```txt
D = 1.60 m, f = 7.50 GHz
d[G] = 20 log10(2) = 6.0206 dB ~ 6.02 dB
[G]  = 39.3936 dBi ~ 39.39 dBi
HPBW = 1.7500 deg ~ 1.75 deg
```

$$\Delta[G] = +6.02\,\text{dB}, \quad [G'] = 39.39\,\text{dBi}, \quad \text{HPBW}' = 1.75^\circ \quad \blacksquare$$

c) With $f' = 2f = 15.00\,\text{GHz}$ and $D = 0.80\,\text{m}$, only the frequency term changes:

$$\Delta[G] = 20\log_{10}\left(\frac{f'}{f}\right) = 20\log_{10}(2) = 6.0206\,\text{dB}$$

$$[G'] = 33.3730 + 6.0206 = 39.3936\,\text{dBi}$$

$$\text{HPBW}' \approx \frac{21}{(15.00)(0.80)} = \frac{21}{12.00} = 1.75^\circ$$

The same physical dish is now twice as large when measured in wavelengths (D/λ doubles), so it behaves exactly like the doubled-diameter antenna in Part B.

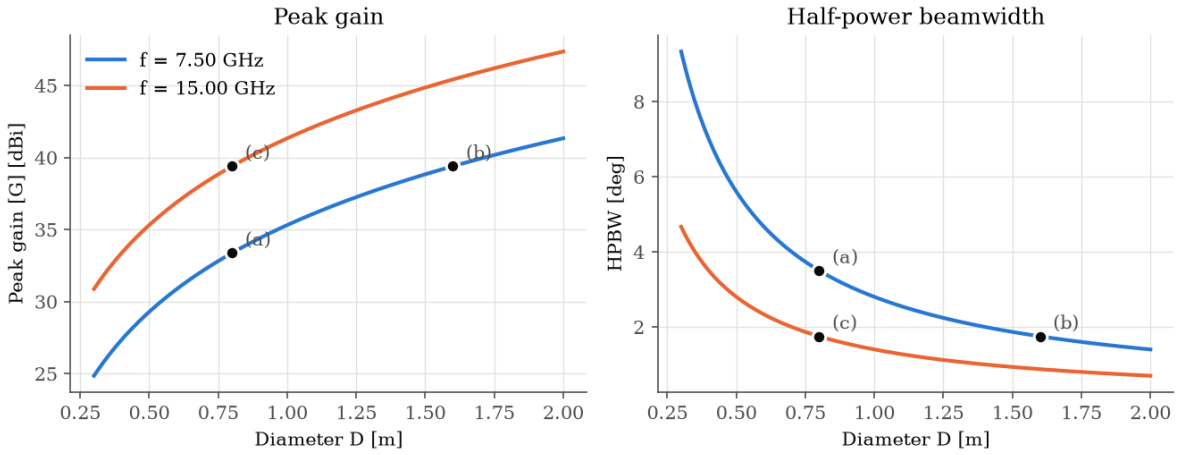


Figure 1: Peak gain and HPBW versus diameter at both frequencies; cases (a)–(c) marked.

This gain increase alone does not establish the change in received power because received power depends on every term of the link equation,

$$[P_r] = [P_t] + [G_t] + [G_r] - [L_{\text{fs}}] - [L_{\text{other}}], \quad [L_{\text{fs}}] = 20\log_{10}\left(\frac{4\pi R f}{c}\right)$$

and several of those also change with frequency:

- **Free-space path loss** rises by the same $20\log_{10}(2) = 6.02\,\text{dB}$ at fixed range. If the other end of the link is a fixed-*gain* antenna (e.g. a low-gain or fixed-beamwidth antenna), the $+6.02\,\text{dB}$ gain is exactly cancelled and $\Delta[P_r] \approx 0\,\text{dB}$; only if the other end is a fixed-*aperture* antenna, whose gain also rises by $6.02\,\text{dB}$, does the link gain a net $+6.02\,\text{dB}$.
- **Pointing loss** grows: the beam is half as wide, so the same pointing error θ costs $4\times$ the loss ($L \propto (\theta/\text{HPBW})^2$).
- **Atmospheric and rain attenuation**, surface-tolerance (Ruze) losses, and receiver noise temperature and hardware losses are generally worse at higher frequency.

So the gain change is only one term; the received power (and E_b/N_0) change must come from the complete link budget.

$$\Delta[G] = +6.02\,\text{dB}, \quad [G'] = 39.39\,\text{dBi}, \quad \text{HPBW}' = 1.75^\circ \quad \blacksquare$$

Problem 2

Pointing and Polarization Loss *(30 pts.)*

A different link has aligned, polarization-matched received power $\left[P_{r_0}\right] = -118.00\,\text{dBW}$. The receiver requires at least $-123.5\,\text{dBW}$. Transmit HPBW is 3.5° with a pointing error of 0.875°. Receive HPBW is 1.4° with a pointing error of 0.35°. Both antennas use linear polarization; their polarization axes differ by 45° about the line of sight. All other losses are already included in $\left[P_{r_0}\right]$.

a) Calculate the transmit and receive pointing losses separately. *(10 pts.)*

b) Calculate polarization loss, actual received power, and margin above the receiver threshold. Does the link meet the threshold? *(10 pts.)*

c) Change only the polarization-axis difference to 60°. Recalculate received power and margin. Explain why correcting polarization alignment can restore this link without changing antenna gain. *(10 pts.)*

Use the following near-boresight approximation and ideal linear-polarization model. Enter θ and full HPBW in the same angular units. Record losses as positive dB values and subtract them once. The pointing approximation does not model sidelobes or nulls.

$$\left[L_{\text{point}}\right] = 12\left(\frac{\theta}{\text{HPBW}}\right)^2$$

$$\left[L_{\text{pol}}\right] = -10\log_{10}(\cos^2(\psi))$$

$$[P_r] = \left[P_{r_0}\right] - \left[L_{\text{pt}}\right] - \left[L_{\text{pr}}\right] - \left[L_{\text{pol}}\right]$$

$$\text{Margin} = [P_r] - \text{Threshold}$$

a) Using the near-boresight approximation with each antenna's own pointing error and full HPBW (both in degrees):

$$\left[L_{\text{pt}}\right] = 12\left(\frac{0.875^\circ}{3.50^\circ}\right)^2 = 12(0.25)^2 = 0.75\,\text{dB}$$

$$\left[L_{\text{pr}}\right] = 12\left(\frac{0.350^\circ}{1.40^\circ}\right)^2 = 12(0.25)^2 = 0.75\,\text{dB}$$

Both errors are one quarter of their beamwidths, so both losses are equal even though the receive beam is narrower.

```txt
[L_pt] = 12 (0.875 / 3.5)^2 = 0.7500 dB
[L_pr] = 12 (0.35 / 1.4)^2 = 0.7500 dB
```

$$\left[L_{\text{pt}}\right] = 0.75\,\text{dB}, \quad \left[L_{\text{pr}}\right] = 0.75\,\text{dB} \quad \blacksquare$$

b) For a $\psi = 45^\circ$ axis difference:

$$\left[L_{\text{pol}}\right] = -10\log_{10}(\cos^2(45^\circ)) = -10\log_{10}(0.5) = 3.0103\,\text{dB}$$

$$\begin{aligned}[P_r] &= \left[P_{r_0}\right] - \left[L_{\text{pt}}\right] - \left[L_{\text{pr}}\right] - \left[L_{\text{pol}}\right] \\ &= -118.00 - 0.75 - 0.75 - 3.0103 = -122.5103\,\text{dBW}\end{aligned}$$

$$\text{Margin} = [P_r] - \text{Threshold} = -122.5103 - (-123.50) = +0.9897\,\text{dB}$$

```txt
psi     = 45 deg
[L_pol] = 3.0103 dB
[P_r]   = -122.5103 dBW ~ -122.51 dBW
Margin  = 0.9897 dB ~ 0.99 dB -> meets threshold
```

The margin is positive, so the link **meets** the receiver threshold, but only by about 1 dB.

$$[L_{\text{pol}}] = 3.01\,\text{dB}, \quad [P_r] = -122.51\,\text{dBW}, \quad \text{Margin} = +0.99\,\text{dB} \quad \blacksquare$$

c) Changing only ψ to 60° (pointing losses unchanged):

$$\begin{aligned} [L_{\text{pol}}] &= -10\log_{10}(\cos^2(60°)) = -10\log_{10}(0.25) = 6.0206\,\text{dB} \\ [P_r] &= -118.00 - 0.75 - 0.75 - 6.0206 = -125.5206\,\text{dBW} \\ \text{Margin} &= -125.5206 - (-123.50) = -2.0206\,\text{dB} \end{aligned}$$

```txt
psi     = 60 deg
[L_pol] = 6.0206 dB
[P_r]   = -125.5206 dBW ~ -125.52 dBW
Margin  = -2.0206 dB ~ -2.02 dB -> fails threshold
```

The margin is negative, so the link **fails** the threshold.

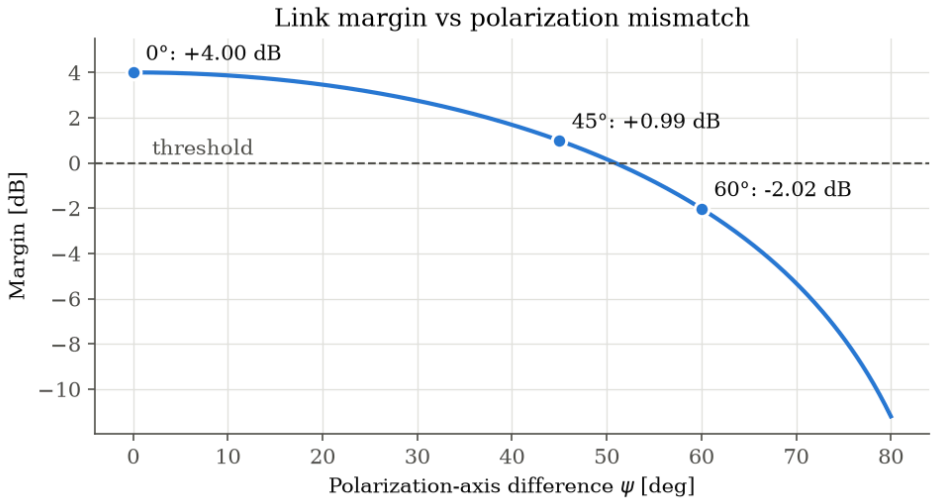


Figure 2: Link margin versus polarization-axis difference ψ (pointing losses fixed at 0.75 dB each).

Polarization loss is a *mismatch* loss: the receive antenna only couples the component of the incident field parallel to its own polarization axis, a fraction $\cos^2\psi$ of the power. It does not depend on either antenna's gain, aperture, or beam pattern, and the full gain is still being delivered along the line of sight. Rotating the feed, the antenna, or the spacecraft about the line of sight to bring $\psi \to 0°$ (or switching both ends to matched circular polarization) drives $[L_{\text{pol}}] \to 0\,\text{dB}$:

$$[P_r] = -118.00 - 0.75 - 0.75 - 0 = -119.50\,\text{dBW}, \quad \text{Margin} = +4.00\,\text{dB}$$

recovering 6.02 dB and restoring the link with no change to antenna gain.

$$[L_{\text{pol}}] = 6.02\,\text{dB}, \quad [P_r] = -125.52\,\text{dBW}, \quad \text{Margin} = -2.02\,\text{dB} \quad \blacksquare$$

Problem 3

Antenna Pointing During a Pass *(20 pts.)*

A spacecraft must keep its science instrument pointed at nadir while communicating continuously throughout a 12-minute overhead pass. Its fixed antenna also points at nadir. At the midpoint (6 minutes), the station lies on antenna boresight: gain is at its peak, range is shortest, and link margin is +5 dB. Toward either end, the station moves off boresight, gain decreases, and range increases. Use the interval approximations below; all other link-budget terms remain unchanged. The link is usable when margin is zero or greater.

Time After Start of Pass	Fixed Antenna Gain Relative to Peak	Additional Path Loss
$(0-2)$ min	-8 dB	4 dB
$(2-5)$ min	-2 dB	1 dB
$(5-7)$ min	0 dB	0 dB
$(7-10)$ min	-2 dB	1 dB
$(10-12)$ min	-8 dB	4 dB

a) For the fixed antenna, calculate margin in each interval. Identify when the link is usable. Does it meet the continuous-communication requirement? *(8 pts.)*

b) Put the same antenna on a gimbal that tracks the station throughout the pass while the spacecraft remains nadir-pointed. Assume ideal tracking, adequate gimbal travel, and no added RF loss: antenna gain stays at its peak, but the additional path loss remains as listed. Calculate margin in each interval. Does this configuration meet the requirement? *(8 pts.)*

c) Explain why a gimbal is needed for this antenna to meet both requirements. Why would rotating the whole spacecraft toward the station not be an acceptable solution? *(4 pts.)*

$$\text{Margin} = 5\,\text{dB} + \text{Gain}_{\text{relative to peak}} - \text{Additional Path Loss}$$

Additional path loss is relative to the pass midpoint. A gimbal changes pointing, not range. No outside research is required.

a) For the fixed, nadir-pointed antenna, $\text{Margin} = 5\,\text{dB} + \text{Gain}_{\text{rel}} - \text{Additional Path Loss}$ in each interval:

$$\begin{aligned}
(0-2)\,\text{min}: &\quad 5 + (-8) - 4 = -7\,\text{dB} \\
(2-5)\,\text{min}: &\quad 5 + (-2) - 1 = +2\,\text{dB} \\
(5-7)\,\text{min}: &\quad 5 + 0 - 0 = +5\,\text{dB} \\
(7-10)\,\text{min}: &\quad 5 + (-2) - 1 = +2\,\text{dB} \\
(10-12)\,\text{min}: &\quad 5 + (-8) - 4 = -7\,\text{dB}
\end{aligned}$$

```
1   Interval  Gain rel. [dB]  Add. loss [dB]  Margin [dB] Usable
2    0-2 min              -8               4           -7     no
3    2-5 min              -2               1            2    yes
4    5-7 min               0               0            5    yes
5   7-10 min              -2               1            2    yes
6  10-12 min              -8               4           -7     no
7
```

```
8  usable time = 8 of 12 min -> NOT continuous
```

The link is usable (margin $\geq 0\,\text{dB}$) only from $(2-10)\,\text{min}$, i.e. 8 min of the 12 min pass. In the first and last 2 min the station is far off boresight and the margin is $-7\,\text{dB}$, so the fixed antenna **does not** meet the continuous-communication requirement.

$$t_{\text{usable}} = (2-10)\,\text{min}\ (8\,\text{min}), \quad \text{Margin}_{\text{min}} = -7\,\text{dB} \quad \blacksquare$$

b) With ideal gimbal tracking the gain stays at peak ($\text{Gain}_{\text{rel}} = 0\,\text{dB}$); only the range-driven path loss remains:

$$\begin{aligned}
(0-2)\,\text{min}: &\quad 5+0-4 = +1\,\text{dB}\\
(2-5)\,\text{min}: &\quad 5+0-1 = +4\,\text{dB}\\
(5-7)\,\text{min}: &\quad 5+0-0 = +5\,\text{dB}\\
(7-10)\,\text{min}: &\quad 5+0-1 = +4\,\text{dB}\\
(10-12)\,\text{min}: &\quad 5+0-4 = +1\,\text{dB}
\end{aligned}$$

```
1   Interval  Gain rel. [dB]  Add. loss [dB]  Margin [dB] Usable                txt
2    0-2 min               0               4           1    yes
3    2-5 min               0               1           4    yes
4    5-7 min               0               0           5    yes
5   7-10 min               0               1           4    yes
6  10-12 min               0               4           1    yes
7
8  usable time = 12 of 12 min -> continuous
```

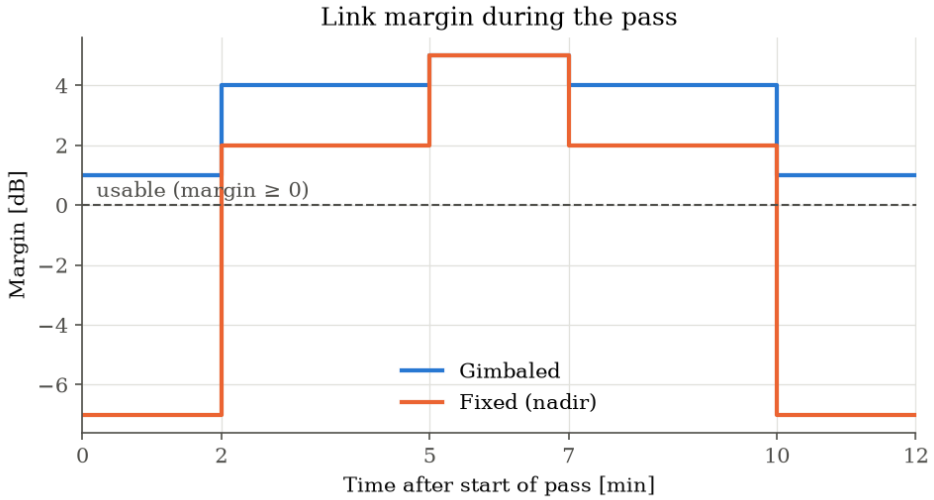


Figure 3: Link margin through the pass for the fixed (nadir) and gimbaled antennas.

Every interval has non-negative margin (minimum $+1\,\text{dB}$ at the ends of the pass), so the gimbaled configuration **meets** the continuous-communication requirement while the spacecraft stays nadir-pointed.

$$\text{Margin}_{\text{min}} = +1\,\text{dB} \geq 0 \Rightarrow \text{continuous over } (0-12)\,\text{min} \quad \blacksquare$$

c) The fixed antenna fails only because of *pointing*: at the ends of the pass the station is off boresight and the $-8\,\mathrm{dB}$ gain loss, combined with the $4\,\mathrm{dB}$ extra path loss, drives the margin to $-7\,\mathrm{dB}$. Range loss alone ($4\,\mathrm{dB}$) fits within the $5\,\mathrm{dB}$ midpoint margin, so the only way to close the link is to keep the antenna boresight on the station. Since the antenna is rigidly aligned with the nadir-pointed body, that requires pointing either the antenna or the whole spacecraft.

Rotating the whole spacecraft toward the station would also rotate the body-fixed science instrument off nadir for most of the pass, which violates the science-pointing requirement (the two requirements need two different pointing directions at the same time). Slewing the bus also costs attitude-control momentum or propellant, settling time and pointing stability, and can disturb thermal and solar-array geometry. A gimbal *decouples* the two pointing directions: the bus stays nadir for science while the antenna tracks the station for communications. A gimbal changes pointing, not range, which is why the residual path loss in Part B remains but is covered by margin.

Gimbal decouples antenna pointing from nadir science pointing. ■

Problem 4

Antenna Mass and Pointing Trade *(20 pts.)*

Continue the spacecraft scenario from Problem 3. The spacecraft must maintain nadir-pointed science observations and communicate throughout the full pass. Both antenna options below have the RF performance used in Problem 3. The maximum installed antenna-system mass is 5 kg.

Option	Complete Installed Mass	Pointing Capability
A: Fixed Antenna	2 kg	Requires spacecraft rotation to point at the station.
B: Gimbaled Antenna	6 kg	Tracks the station while the spacecraft maintains its science attitude.

The stated masses include the aperture, feed, supports, and any deployment or pointing mechanisms. These are supplied design assumptions, not specifications for a named mission.

a) Using your results from Problem 3, identify which option meets the communications and science requirements. Compare each option with the mass limit. Does either meet all requirements as proposed? *(8 pts.)*

b) Recommend one design or requirement change that makes the mission feasible. State its cost or consequence, including any mass reallocation, hardware redesign, or reduction in communications or science performance. *(6 pts.)*

c) Explain why antenna mass location matters to spacecraft slewing and pointing, even when two designs have the same total mass. *(6 pts.)*

a) Using the Problem 3 results and the 5.0 kg limit:

Option	Comms (full pass)	Science (nadir)	Mass vs. limit	All met?
A: Fixed	Only if the spacecraft slews (fails if nadir: -7 dB at the pass ends)	Only if it does *not* slew	2.0 kg $\leq$ 5.0 kg (3.0 kg under)	No
B: Gimbaled	Yes ($\text{Margin}_{\text{min}} = +1$ dB)	Yes	6.0 kg $>$ 5.0 kg (1.0 kg, 20 % over)	No

```
1       Option  Mass [kg] Comms (nadir)         Science  Excess [kg] Mass OK
2     A: Fixed        2.0            no only if no slew         -3.0     yes
3  B: Gimbaled        6.0           yes             yes          1.0      no
4
5  mass limit = 5.0 kg; B exceeds by 1.0 kg (20% of the limit)
```

Option A fits the mass limit but cannot satisfy communications and science at the same time: nadir-pointed it has only 8 min of usable link, and slewing to close the link breaks the science attitude. Option B meets both the communications and science requirements but exceeds the mass limit by 1.0 kg. **Neither option meets all requirements as proposed.**

$$m_B - m_{\text{max}} = 6.0 - 5.0 = 1.0 \text{ kg over} \quad \blacksquare$$

A cannot meet comms and science together.

b) **Recommendation:** select Option B (gimbaled) and raise the antenna-system mass allocation from 5.0 kg to 6.0 kg by reallocating 1.0 kg from the spacecraft's system-level mass margin (or from another subsystem's allocation). This is the only option that keeps both full-pass communications and continuous nadir science with no performance reduction, and the shortfall is a single, modest 1.0 kg.

Cost and consequences:

- Mass reallocation: system mass margin drops by 1.0 kg, leaving less reserve for later growth, or another subsystem must be lightened or descoped (e.g. less propellant, structure or battery mass), which may itself cost lifetime or capability. The change must be approved through the mass budget/requirements process.
- Hardware: a gimbal adds a mechanism (actuators, drive electronics, harness, cable wrap or rotary joint), a single-point-failure risk, power consumption, extra test and qualification, and cost.
- Attitude control: gimbal motion produces reaction torques and jitter that the ACS must absorb while holding nadir.
- Performance: with ideal tracking, communications keep at least +1 dB margin through the pass, and science pointing is unaffected.

$$m_{\text{alloc}'} = 5.0\,\text{kg} + 1.0\,\text{kg} = 6.0\,\text{kg} \geq m_B \quad \blacksquare$$

c) Slewing and pointing depend on the spacecraft's *inertia* and *center of mass*, not just its total mass. For a mass m at distance r from the rotation axis, the moment of inertia is

$$I = \sum_i m_i r_i^2, \quad \tau = I\alpha, \quad H = I\omega$$

so the same 6.0 kg placed on a boom 1 m from the center of mass adds about $6\,\text{kg}\,\text{m}^2$, while the same mass mounted close to the center adds almost nothing. Consequently:

- **Slewing:** larger I requires more torque for the same angular acceleration, or more time and stored momentum ($H = I\omega$) for the same slew rate, driving larger reaction wheels, more momentum dumping, and more power or propellant.
- **Center-of-mass offset:** an off-axis antenna shifts the center of mass, adding disturbance torques (gravity gradient, solar pressure, drag, thruster misalignment) that the ACS must counter continuously.
- **Cross-coupling:** asymmetric mass creates products of inertia, so a commanded rotation about one axis induces motion about others.
- **Moving mass and flexibility:** a gimbaled or boom-mounted antenna moves mass relative to the bus, producing reaction torques and lower-frequency structural modes that cause jitter and degrade the fine pointing of both the antenna and the science instrument.

Two designs with equal total mass can therefore have very different slew times, actuator sizing, and pointing stability.

$I = \sum_i m_i r_i^2$ depends on mass location, not just total mass. ■

Code

./src/index.py Python

```python
import os

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd

# repository root (./submission.py is a symlink to ./src/index.py)
ROOT = os.path.dirname(os.path.abspath(__file__))
if os.path.basename(ROOT) == "src":
    ROOT = os.path.dirname(ROOT)

# plot styling
SERIES = ["#2a78d6", "#eb6834", "#1baf7a"]
INK = "#52514e"
plt.rcParams.update(
    {
        "figure.dpi": 150,
        "font.family": "serif",
        "axes.spines.top": False,
        "axes.spines.right": False,
        "axes.edgecolor": INK,
        "axes.labelcolor": "#0b0b0b",
        "axes.grid": True,
        "grid.color": "#e4e3df",
        "grid.linewidth": 0.6,
        "xtick.color": INK,
        "ytick.color": INK,
        "lines.linewidth": 2,
    }
)


def heading(problem):
    print(f"\n------\n {problem}\n------")


def save_text(name, text):
    print(text)
    with open(os.path.join(ROOT, "outputs/text", f"{name}.txt"), "w", encoding="utf-8") as f:
        f.write(text + "\n")


def save_fig(fig, name):
    fig.tight_layout()
```

```
45      fig.savefig(os.path.join(ROOT, "outputs/figures", f"{name}.png"))
46      plt.close(fig)
47
48
49  # p01: antenna gain [dBi] and full HPBW [deg] (D in m, f in GHz)
50  def gain_db(D, f):
51      return 17.81 + 20 * np.log10(D) + 20 * np.log10(f)
52
53
54  def hpbw(D, f):
55      return 21 / (f * D)
56
57
58  # p02: pointing and polarization losses [dB]
59  def pointing_loss(theta, bw):
60      return 12 * (theta / bw) ** 2
61
62
63  def pol_loss(psi_deg):
64      return -10 * np.log10(np.cos(np.radians(psi_deg)) ** 2)
65
66
67  def main():
68      os.makedirs(os.path.join(ROOT, "outputs/text"), exist_ok=True)
69      os.makedirs(os.path.join(ROOT, "outputs/figures"), exist_ok=True)
70
71      #########
72      # p01 #
73      #########
74      heading("p01")
75
76      D0, f0 = 0.80, 7.50
77
78      # a) original antenna
79      G_a, bw_a = gain_db(D0, f0), hpbw(D0, f0)
80      save_text(
81          "s01a",
82          f"D = {D0:.2f} m, f = {f0:.2f} GHz\n"
83          f"20 log10(D) = {20 * np.log10(D0):.4f} dB\n"
84          f"20 log10(f) = {20 * np.log10(f0):.4f} dB\n"
85          f"[G]  = {G_a:.4f} dBi ~ {G_a:.2f} dBi\n"
86          f"HPBW = {bw_a:.4f} deg ~ {bw_a:.2f} deg",
87      )
88
89      # b) doubled diameter
90      G_b, bw_b = gain_db(2 * D0, f0), hpbw(2 * D0, f0)
```

```
91      save_text(
92          "s01b",
93          f"D = {2 * D0:.2f} m, f = {f0:.2f} GHz\n"
94          f"d[G] = 20 log10(2) = {G_b - G_a:.4f} dB ~ {G_b - G_a:.2f} dB\n"
95          f"[G]  = {G_b:.4f} dBi ~ {G_b:.2f} dBi\n"
96          f"HPBW = {bw_b:.4f} deg ~ {bw_b:.2f} deg",
97      )
98
99      # c) doubled frequency
100     G_c, bw_c = gain_db(D0, 2 * f0), hpbw(D0, 2 * f0)
101     save_text(
102         "s01c",
103         f"D = {D0:.2f} m, f = {2 * f0:.2f} GHz\n"
104         f"d[G] = 20 log10(2) = {G_c - G_a:.4f} dB ~ {G_c - G_a:.2f} dB\n"
105         f"[G]  = {G_c:.4f} dBi ~ {G_c:.2f} dBi\n"
106         f"HPBW = {bw_c:.4f} deg ~ {bw_c:.2f} deg\n"
107         f"d[L_fs] = 20 log10(2) = {20 * np.log10(2):.4f} dB (free-space loss also rises)",
108     )
109
110     # gain and HPBW vs diameter (two charts, one axis each)
111     D = np.linspace(0.3, 2.0, 200)
112     fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(9, 3.6))
113     for i, f in enumerate([f0, 2 * f0]):
114         ax1.plot(D, gain_db(D, f), color=SERIES[i], label=f"f = {f:.2f} GHz")
115         ax2.plot(D, hpbw(D, f), color=SERIES[i], label=f"f = {f:.2f} GHz")
116     cases = [("a", D0, f0), ("b", 2 * D0, f0), ("c", D0, 2 * f0)]
117     for name, Dc, fc in cases:
118         for ax, y in [(ax1, gain_db(Dc, fc)), (ax2, hpbw(Dc, fc))]:
119             ax.plot(Dc, y, "o", ms=7, color="#0b0b0b", mec="white", mew=1.5, zorder=3)
120             ax.annotate(f"({name})", (Dc, y), xytext=(7, 4), textcoords="offset points",
                color=INK)
121     ax1.set(xlabel="Diameter D [m]", ylabel="Peak gain [G] [dBi]", title="Peak gain")
122     ax2.set(xlabel="Diameter D [m]", ylabel="HPBW [deg]", title="Half-power beamwidth")
123     ax1.legend(frameon=False)
124     save_fig(fig, "s01")
125
126     #########
127     # p02 #
128     #########
129     heading("p02")
130
131     Pr0, thresh = -118.00, -123.50
132     bw_t, th_t = 3.50, 0.875
133     bw_r, th_r = 1.40, 0.350
134
135     # a) pointing losses
136     L_pt, L_pr = pointing_loss(th_t, bw_t), pointing_loss(th_r, bw_r)
```

```
137     save_text(
138         "s02a",
139         f"[L_pt] = 12 ({th_t} / {bw_t})^2 = {L_pt:.4f} dB\n"
140         f"[L_pr] = 12 ({th_r} / {bw_r})^2 = {L_pr:.4f} dB",
141     )
142
143     # b), c) polarization loss, received power, margin
144     def link(psi):
145         L_pol = pol_loss(psi)
146         Pr = Pr0 - L_pt - L_pr - L_pol
147         return L_pol, Pr, Pr - thresh
148
149     for part, psi in [("b", 45), ("c", 60)]:
150         L_pol, Pr, M = link(psi)
151         save_text(
152             f"s02{part}",
153             f"psi     = {psi} deg\n"
154             f"[L_pol] = {L_pol:.4f} dB\n"
155             f"[P_r]   = {Pr:.4f} dBW ~ {Pr:.2f} dBW\n"
156             f"Margin  = {M:.4f} dB ~ {M:.2f} dB -> {'meets' if M >= 0 else 'fails'}
                threshold",
157         )
158     _L_pol0, Pr0_al, M0 = link(0)
159     print(f"aligned (psi = 0): [P_r] = {Pr0_al:.2f} dBW, Margin = {M0:.2f} dB")
160
161     # margin vs polarization mismatch
162     psi = np.linspace(0, 80, 400)
163     fig, ax = plt.subplots(figsize=(6.5, 3.6))
164     ax.plot(psi, link(psi)[2], color=SERIES[0])
165     ax.axhline(0, color=INK, lw=1, ls="--")
166     ax.annotate("threshold", (2, 0), xytext=(0, 4), textcoords="offset points", color=INK)
167     for p in [0, 45, 60]:
168         M = link(p)[2]
169         ax.plot(p, M, "o", ms=7, color=SERIES[0], mec="white", mew=1.5, zorder=3)
170         ax.annotate(f"{p}°: {M:+.2f} dB", (p, M), xytext=(6, 6), textcoords="offset
            points", color="#0b0b0b")
171     ax.set_ylim(top=5.5)
172     ax.set(xlabel=r"Polarization-axis difference $\psi$ [deg]", ylabel="Margin [dB]",
        title="Link margin vs polarization mismatch")
173     save_fig(fig, "s02c")
174
175     #########
176     # p03 #
177     #########
178     heading("p03")
179
180     intervals = pd.DataFrame(
```

```
        {
            "start": [0, 2, 5, 7, 10],
            "end": [2, 5, 7, 10, 12],
            "gain_rel": [-8, -2, 0, -2, -8],
            "path_loss": [4, 1, 0, 1, 4],
        }
    )
    intervals["interval"] = [f"{s}-{e} min" for s, e in zip(intervals.start, intervals.end)]

    # a) fixed antenna; b) gimbaled antenna (gain held at peak)
    intervals["fixed"] = 5 + intervals.gain_rel - intervals.path_loss
    intervals["gimbal"] = 5 + 0 - intervals.path_loss

    for part, col in [("a", "fixed"), ("b", "gimbal")]:
        tab = intervals[["interval", "gain_rel", "path_loss", col]].copy()
        if col == "gimbal":
            tab["gain_rel"] = 0
        tab["usable"] = np.where(tab[col] >= 0, "yes", "no")
        tab.columns = ["Interval", "Gain rel. [dB]", "Add. loss [dB]", "Margin [dB]", "Usable"]
        usable = (intervals[col] >= 0) * (intervals.end - intervals.start)
        save_text(
            f"s03{part}",
            tab.to_string(index=False)
            + f"\n\nusable time = {usable.sum()} of 12 min -> "
            + ("continuous" if usable.sum() == 12 else "NOT continuous"),
        )

    fig, ax = plt.subplots(figsize=(6.5, 3.6))
    edges = np.r_[intervals.start, 12]
    for i, (col, lbl) in enumerate([("gimbal", "Gimbaled"), ("fixed", "Fixed (nadir)")]):
        ax.step(edges, np.r_[intervals[col], intervals[col].iloc[-1]], where="post", color=SERIES[i], label=lbl)
    ax.axhline(0, color=INK, lw=1, ls="--")
    ax.annotate("usable (margin ≥ 0)", (0.2, 0), xytext=(0, 4), textcoords="offset points", color=INK)
    ax.set(xlabel="Time after start of pass [min]", ylabel="Margin [dB]", title="Link margin during the pass", xticks=edges, xlim=(0, 12))
    ax.legend(frameon=False, loc="lower center")
    save_fig(fig, "s03")

    #########
    # p04 #
    #########
    heading("p04")

```

```
223     m_max = 5.0
224     options = pd.DataFrame(
225         {
226             "Option": ["A: Fixed", "B: Gimbaled"],
227             "Mass [kg]": [2.0, 6.0],
228             "Comms (nadir)": ["no", "yes"],
229             "Science": ["only if no slew", "yes"],
230         }
231     )
232     options["Excess [kg]"] = options["Mass [kg]"] - m_max
233     options["Mass OK"] = np.where(options["Mass [kg]"] <= m_max, "yes", "no")
234     save_text(
235         "s04a",
236         options.to_string(index=False)
237         + f"\n\nmass limit = {m_max:.1f} kg; B exceeds by {options['Excess [kg]'][1]:.1f}
            kg"
238         + f" ({100 * options['Excess [kg]'][1] / m_max:.0f}% of the limit)",
239     )
240
241
242 if __name__ == "__main__":
243     main()
```